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Population genetic structure and comparative diversity of smallmouth bass *Micropterus dolomieu*: congruent patterns from two genomes

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Genetic diversity and divergence patterns of smallmouth bass *Micropterus dolomieu* spawning groups are analysed across its northern native range with mtDNA cytochrome *b* gene sequences and eight unlinked nuclear DNA microsatellite loci. Results reveal high levels of genetic variability and significant differences in allelic representation among populations (mtDNA: mean ± s.e., $H_D = 0{\cdot}50 \pm 0{\cdot}06$, mean ± s.e., $\theta_{ST} = 0{\cdot}41 \pm 0{\cdot}02$ and microsatellites: mean ± s.e. $H_O = 0{\cdot}46 \pm 0{\cdot}03$, mean ± s.e. $\theta_{ST} = 0{\cdot}25 \pm 0{\cdot}01$). The distributions of 28 variant mtDNA haplotypes, which differ by an average of 3·94 nucleotides (range = 1–8), denote divergent representation among geographic areas. Microsatellite data support nine primary population groups, whose high self-assignment probabilities likewise display marked divergence. Genetic patterns demonstrate: (1) high genetic diversity in both genomes, (2) significant divergence among populations, probably resulting from natal site homing and low lifetime migration, (3) support for three post-glacial refugia that variously contributed to the current northern populations, which remain evident today despite waterway connectivity and (4) a weak yet significant genetic isolation by geographic distance pattern, indicating that other processes affect the differences among populations, such as territoriality and site fidelity.

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Key words: cytochrome *b*; Great Lakes; *Micropterus dolomieu*; microsatellites; phylogeography; smallmouth bass.

INTRODUCTION

Wide-ranging species often harbour a large number of genetically variable populations, whose broad-scale genetic patterns reflect their demographic and geographic histories and interactions (Slatkin, 1987; Avise, 2000; Manel *et al.*, 2003). Fine-scale population structure of such species is influenced by gene flow and differentiation patterns that are regulated by spatial patterns of habitat availability, connectivity among locations, dispersal ability and reproductive behaviour. Population divergence

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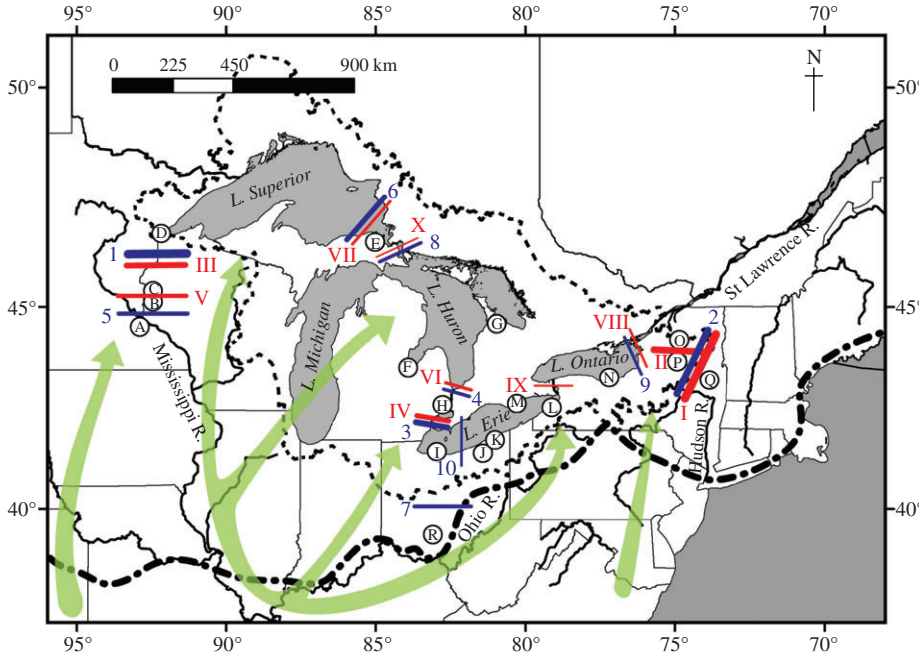

FIG. 1. Hypothesized colonization pathway (↑) and sampling sites (Ⓐ-----, see Table I for names) for 18 *Micropterus dolomieu* populations. Ⓡ, The extent of the Great Lakes catchment; ------, the maximum extent of the Pleistocene glaciations; ═══, inferred barriers to gene flow based on mtDNA (——, ranked 1–10), and eight microsatellite (——, ranked I–XI) from BARRIER analyses (Manni *et al.*, 2004). Thicker lines indicate greater relative support for divergence between populations separated by the barrier. Relative support for each microsatellite barrier (rank, bootstrap support, *N* supporting loci) was: I = 0·694, 6; II = 0·637, 6; III = 0·604, 7; IV = 0·734, 6; V = 0·684, 5; VI = 0·689, 7; VII = 0·658, 6; VIII = 0·605, 5; IX = 0·591, 5; X = 0·648, 6.

decreases with increased gene flow, facilitated by higher degrees of habitat connectivity and dispersal (Boizard *et al.*, 2009; Fitzpatrick *et al.*, 2014; Kutschera *et al.*, 2016). In contrast, habitat isolation and behavioural site fidelity will reduce gene flow and increase population differences (Lindsay *et al.*, 2008; Nichols *et al.*, 2012; Baker, C.S. *et al.*, 2013; Baker, W. H. *et al.*, 2013). Furthermore, compared with large populations, small or isolated ones often contain lower genetic diversity resulting from genetic drift, population bottlenecks and localized selection (Nei *et al.*, 1975; Hedrick & Kalinowski, 2000; Vandewoestijne *et al.*, 2008; Bouzat, 2010; Puckett *et al.*, 2014).

Broad-scale population genetic patterns generally are shaped by historical climatic and geographic influences on demography, including barriers to dispersal and connectivity. In temperate North America, the primary historic determinant was the Pleistocene glaciations from *c.* 2 580 000 to *c.* 12 000 years ago (ya), which markedly limited available habitat (Fig. 1; Hewitt, 2004; Gibbard *et al.*, 2010). Populations of freshwater fishes were concentrated in southerly glacial refugia, where they experienced isolation and bottlenecks, followed by subsequent founder effects with recolonization of northern habitats after glacial recession (Bernatchez, 1997; Hewitt, 2000; Gugerli *et al.*, 2001). Interestingly, recent studies have indicated a

seat of high population genetic variability in several fish species in the Laurentian Great Lakes, which offered huge and abundant new habitat formed by the glaciers. This high diversity is exemplified in today's Great Lakes' populations of smallmouth bass *Micropterus dolomieu* Lacépède 1802 (Stepien *et al.*, 2007), yellow perch *Perca flavescens* (Mitchill 1814) (Sepulveda-Villet & Stepien, 2012; Sullivan & Stepien, 2014, 2015; Stepien *et al.*, 2015*a*, *b*), walleye *Sander vitreus* (Mitchill 1818) (Stepien *et al.*, 2009, 2015*b*; Haponski & Stepien, 2014), lake whitefish *Coregonus clupeaformis* (Mitchill 1818) (Bernatchez & Dodson, 1991) and brown bullhead *Ameiurus nebulosus* (LeSueur 1819) (Murdoch & Hebert, 1997). The pattern of substantial genetic diversity in recolonized areas is believed to reflect admixture of populations from several different glacial refugia (Bernatchez & Dodson, 1991; Stepien *et al.*, 2007, 2015*a*, *b*; Sepulveda-Villet & Stepien, 2012).

The genus *Micropterus* Lacépède 1802 (family Centrarchidae) contains 13 North American species of black basses (Baker, C.S. *et al.*, 2013; Baker, W. H. *et al.*, 2013), which are top carnivores and popular sport fishes (Bagley *et al.*, 2011). The wide-ranging smallmouth bass *M. dolomieu* possesses the most northerly range, which encompasses the entirety of the Great Lakes, along with adjacent rivers and streams (Bagley *et al.*, 2011). This broad range provides an interesting case study for investigating biogeographic patterns in formerly glaciated regions and contemporary influences on these patterns. The present study analyses the relative diversity and divergence from both the mitochondrial (mt) and nuclear genomes, with the aim of elucidating the interplay between historic and contemporary factors on population genetic structure. Since mtDNA is maternally inherited and largely free of recombination, its patterns of genetic variation are more conserved, prone to bottlenecks and often retain the evidence of historic differentiation at a broader geographic scale. In contrast, nuclear DNA microsatellites are bi-parentally inherited, have a higher mutation rate and may better reflect contemporary population genetic relationships at finer temporal and spatial scales.

An earlier study by Stepien *et al.* (2007) evaluated eight microsatellite loci of *M. dolomieu* populations across North America, discerning pronounced divergences and primary genetic barriers separating the Hudson River, Ohio River and upper Mississippi River population groups. Stepien *et al.* (2007) noted that its historical glacial refugia signatures were similar to patterns delineated in other species, including *S. vitreus* (Stepien & Faber, 1998; Stepien *et al.*, 2009), *A. nebulosus* (Murdoch & Hebert, 1997) and *P. flavescens* (Sepulveda-Villet & Stepien, 2012). Here, new analyses of those results are compared and contrasted with a new mtDNA cytochrome *b* (*cytb*) sequence data set to provide a more complete view of *M. dolomieu* population genetic patterns and colonization history across the Great Lakes and adjacent river systems.

The project aim was to identify the factors that shaped the population genetic patterns observed today for *M. dolomieu*, comparing and contrasting results from two genomes. Similarities and differences of genetic and geographic factors from mtDNA *cytb* sequences and eight nuclear microsatellites are investigated for 18 *M. dolomieu* populations across once-glaciated areas (Fig. 1). The following questions are addressed: are similar levels of genetic diversity found among *M. dolomieu* populations in both genomes; are their patterns of genetic divergence congruent; is genetic variation significantly partitioned among watersheds, lakes, rivers and basins;

do patterns of differentiation reflect isolation by geographic distance, measured by nearest waterway connectivity?

MATERIALS AND METHODS

SAMPLE COLLECTION, DNA EXTRACTION, AMPLIFICATION AND SEQUENCING

Spawning-condition adult *M. dolomieu* were assayed from 18 native population locations, avoiding areas that had been stocked and surveying a wide variety of habitats in the respective locales (Table I). Collections were made by federal, state and provincial agencies, colleagues and laboratory members (under their respective regulations and permits), with those by the Stepien Laboratory under permits issued from the Michigan Department of Natural Resources and Ohio Division of Wildlife (#140160), with the University of Toledo approved Institutional Animal Care and Use Committee (IACUC) protocol (#105400). Fin-clip tissues were preserved in 95% ethanol in the field and stored at room temperature in the Stepien Laboratory. Genomic DNA was extracted and purified using DNeasy Qiaquick kits (Qiagen; www.qiagen.com). Aliquots were labelled, frozen at −80° C and archived.

The entire mtDNA cytochrome oxidase subunit *b* (*cytb*) gene (1140 bp) was sequenced and analysed from 225 individuals, which were a representative subset of 357 individuals whose nuclear microsatellite variation was the focus of Stepien *et al.* (2007). The *cytb* gene was amplified using the polymerase chain reaction (PCR) and sequenced in both directions using the primers *Song-F* and *Song-R* (Song *et al.*, 1998), *Basscytbf-1* (Near *et al.*, 2003) and *Smb-2r* (5′-CCCTTAGTAACTCCGCCACAC-3′; developed by the Stepien Laboratory). Sequencing reactions were performed at the Cornell University Life Sciences Core Laboratories Center (http://www.biotech.cornell.edu/biotechnology-resource-center-brc/), which used automated Applied Biosystems 3730 (ABI; www.appliedbiosystems.com) with big dye terminator chemistry sequencing. The sequences were then aligned with BIOEDIT 7.25 (http://www.mbio.ncsu.edu/bioedit/bioedit.html/; Hall, 1999). Unique haplotypes were identified and compared with Stepien Laboratory databases and other *M. dolomieu* sequences (Roe *et al.*, 2002; Near *et al.*, 2003) from GenBank (http://www.ncbi.nlm.nih.gov/Genbank/). Accession numbers for haplotype sequences in the present study are: 1 = KU171303, 2 = KU171304, 3 = KU171305, 4 = KU171306, 5 = KU171307, 6 = KU171308, 7 = KU171309, 8 = KU171310, 9 = KU171311, 10 = KU171312, 11 = KU171313, 12 = KU171314, 13 = KU171315, 14 = KU171316, 15 = KU171317, 16 = KU171318, 17 = KU171319, 18 = KU171320, 19 = KU171321, 20 = KU171322, 21 = KU171323, 22 = KU171324, 23 = KU171325, 24 = KU171326, 25 = KU171327, 26 = KU171328, 27 = KU171329 and 28 = KU171330.

Eight nuclear microsatellite DNA loci: Mdo-2, Mdo-8, Mdo-9 (Malloy *et al.*, 2000), Mdo-3, Mdo-5, Mdo-11 (Malloy *et al.*, 2000; Coughlin *et al.*, 2003), RB-7 (DeWoody *et al.*, 1998) and MS-19 (DeWoody *et al.*, 2000) were reanalysed from populations reported in Stepien *et al.* (2007).

MITOCHONDRIAL DNA CYTOCHROME *B* DATA ANALYSIS

Phylogenetic relationships among mtDNA *cytb* haplotypes were analysed with MRBAYES 3.2.2 using a metropolis-coupled Markov chain Monte Carlo-(MCMC) approach (Huelsenbeck & Ronquist, 2001; Ronquist *et al.*, 2011; http://mrbayes.sourceforge.net/). The corrected Akaike information criterion (AICc) from jMODELTEST 2 (Guindon & Gascuel, 2003; Darriba *et al.*, 2012; https://github.com/ddarriba/jmodeltest2/) was employed to assess the most appropriate model of substitution, which selected the Tamura & Nei (1993) model, with unequal frequencies and invariable sites ($I = 0{\cdot}893$). Analyses were run for 10 million generations, with sampling every 100 iterations. Metropolis-coupled MCMC burn-in used log-likelihood values per generation to identify when stationarity was reached; 25% of the generations were discarded, along with trees and values prior to burn-in. A 50% majority-rule consensus tree was based

TABLE I. Genetic diversity of *Micropterus dolomieu* populations based on mtDNA *cytb* sequences and eight nuclear DNA microsatellite loci. Sampling locations correspond to sites A–R in Fig. 1

			mtDNA haplotypes					Microsatellite loci									
Locality	Latitude (° N)	Longitude (° W)	N	N_H	H_D	N_{PH}	p_{PH}	N	N_A	H_O	H_E	N_{PA}	p_{PA}	Sibs	p_S	F_{IS}	IR
Upper Mississippi River	–	–	28	5	0·22	3	0·60	28	45	0·54	0·63	4	0·09	0	0	0·04	0·47
A. Cannon River, MN	44·52	92·88	12	5	0·67	3	0·60	12	31	0·57	0·57	1	0·03	0	0	−0·01	0·42
B. Apple River, WI	45·38	92·44	9	1	0	0	0	9	27	0·49	0·51	1	0·04	0	0	0·04	0·50
C. St·Croix River, MN	45·13	92·44	7	1	0	0	0	7	26	0·55	0·69	2	0·08	0	0	0·08	0·48
Lake Superior	–	–	9	3	0·50	0	0	9	27	0·47	0·54	0	0	2	0·22	0·04	0·52
D. St Louis Bay, MN	46·72	92·16	6	2	0·33	0	0	6	24	0·60	0·54	0	0	0	0	−0·11	0·46
E. Whitefish Bay, MI	46·52	85·01	3	2	0·67	0	0	3	14	0·33	0·34	0	0	2	0·67	0·03	0·58
Lake Huron	–	–	27	9	0·77	4	0·44	42	31	0·49	0·41	0	0	2	0·05	−0·07	0·52
F. Saginaw Bay, MI	43·55	83·90	7	4	0·71	1	0·25	7	23	0·59	0·50	0	0	0	0	−0·17	0·41
G. Georgian Bay, ON	44·59	80·94	20	6	0·83	3	0·50	35	29	0·38	0·39	0	0	2	0·06	0·04	0·62
Lake St Clair	–	–	18	3	0·22	1	0·33	31	32	0·49	0·58	1	0·03	4	0·13	0·15	0·57
H. Anchor Bay, MI	42·63	82·77	18	3	0·22	1	0·33	31	32	0·49	0·58	1	0·03	4	0·13	0·15	0·57
Lake Erie	–	–	76	13	0·67	7	0·54	135	45	0·49	0·50	2	0·04	24	0·18	−0·01	0·51

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TABLE I. continued

Locality	Latitude (° N)	Longitude (° W)	mtDNA haplotypes					Microsatellite loci									
			N	N_H	H_D	N_{PH}	p_{PH}	N	N_A	H_O	H_E	N_{PA}	p_{PA}	Sibs	p_S	F_{IS}	IR
I. Sandusky Bay, OH	41·45	82·96	15	5	0·70	2	0·40	20	34	0·54	0·54	0	0	0	0	0·01	0·46
J. Chagrin River, OH	41·41	81·41	11	3	0·56	1	0·33	11	25	0·56	0·51	2	0·08	0	0	−0·10	0·43
K. Grand River, OH	41·75	80·99	18	7	0·85	3	0·43	26	29	0·51	0·51	0	0	7	0·27	−0·01	0·54
Lake Cattaraugus Creek, NY	42·57	79·11	16	5	0·65	1	0·20	30	28	0·47	0·48	0	0	0	0	0·03	0·54
M. Long Point Bay, ON	42·66	80·26	16	3	0·58	0	0	48	31	0·38	0·39	0	0	17	0·35	0·03	0·59
Lake Ontario	–	–	20	3	0·58	0	0	20	25	0·38	0·51	0	0	0	0	0·26	0·62
N. Fairbanks Point, NY	43·30	77·18	20	3	0·58	0	0	20	25	0·38	0·51	0	0	0	0	0·26	0·62
St Lawrence R·	–	–	24	5	0·39	1	0·20	59	26	0·28	0·39	0	0	26	0·44	0·03	0·72
O. Cranberry Lake, NY	44·21	74·84	12	3	0·62	1	0·33	29	25	0·39	0·39	0	0	6	0·21	0·02	0·61
P. Little Moose Lake, NY	43·69	74·92	12	2	0·17	0	0	30	16	0·16	0·17	0	0	20	0·67	0·03	0·83
Hudson River	–	–	10	3	0·38	0	0	16	21	0·32	0·36	0	0	5	0·31	0·10	0·68
Q. Hudson River, NY	43·24	73·83	10	3	0·38	0	0	16	21	0·32	0·35	0	0	5	0·31	0·10	0·68
Ohio River	–	–	16	4	0·52	1	0·25	17	38	0·59	0·61	1	0·03	0	0	0·04	0·41
R. Paint Creek, OH	39·32	83·08	16	4	0·52	1	0·25	17	38	0·59	0·61	1	0·03	0	0	0·04	0·41
Mean			12·6	3·4	0·50	0·9	0·20	19·8	26·6	0·46	0·48	0·44	0·02	3·5	0·15	0·02	0·54

F_{IS}, deviation from Hardy–Weinberg equilibrium within subpopulations, H_D, haplotypic diversity; H_E, expected heterozygosity; H_O, observed heterozygosity; IR, internal relatedness; N, number of individuals; N_A, number of alleles; N_H, number of haplotypes; N_{PA}, number of private alleles; N_{PH}, number of private haplotypes; p_S, proportion of full sibs in a population sample; p_{PA}, proportion of private alleles; p_{PH}, proportion of private haplotypes; Sibs, the number of full siblings in the population sample.

on the remaining generations and branch support was obtained *via* posterior probability distribution (Holder & Lewis, 2003). The tree was rooted to the putative sister species spotted bass *Micropterus punctulatus* (Rafinesque 1819) (Johnson *et al.*, 2001; Near *et al.*, 2004, 2005; Bagley *et al.*, 2011), using GenBank #HM070928.1 from Bagley *et al.* (2011). Relationships among the haplotypes were further depicted and compared using a statistical parsimony haplotype network in PopART (http://popart.otago.ac.nz/index.shtml/; Leigh & Bryant, 2015).

MICROSATELLITE DNA DATA ANALYSIS

Genotype frequencies for previously collected microsatellite data by Stepien *et al.* (2007) were re-evaluated for conformance to Hardy–Weinberg equilibrium (HWE) expectations at each locus, with significance estimated with GENEPOP using MCMC and 1000 randomizations (Guo & Thompson, 1992). Deviations from HWE were tested for heterozygote deficiency and null (non-amplified) alleles in MICRO-CHECKER 2.2.3 (http://www.norwichresearchpark.com/research1/researchgroups/elsa/software/microchecker.aspx/; van Oosterhout *et al.*, 2004). Genotype frequencies also were analysed for evidence of linkage disequilibrium (LD). Significance levels for HWE and LD were adjusted using sequential Bonferroni correction (Rice, 1989). LOSITAN (http://popgen.net/soft/lositan/; Beaumont & Nichols, 1996; Antao *et al.*, 2008) was used to evaluate whether loci were under possible selection.

Numbers of alleles (N_A) and allelic richness (A_R; number of alleles per locus, adjusted for sample size using rarefaction following El Mousadik & Petit (1996)) were calculated with FSTAT. Numbers of full-siblings were estimated with COLONY 2.0.5.0 (http://www.zsl.org/science/software/colony/; Jones & Wang, 2009). Levels of relatedness further were evaluated by computing population F_{IS} values in GENEPOP and by computing internal relatedness (IR; Amos *et al.*, 2001).

Demographic partitioning of genetic variation was examined using STRUCTURE 2.3.4 (http://pritchardlab.stanford.edu/structure.html/; Pritchard *et al.*, 2000; Pritchard & Wen, 2004), which assessed membership of individuals to $K = 1$ to $K = 18$ possible population groups (up to the total N of populations), independent of sampling location identity. Results were analysed using 10 independent runs for each, with burn-ins of 100 000 and 500 000 MCMC replicates, an admixture model, initial inferred $\alpha = 1{\cdot}0$, assumption of correlated alleles, prior F_{ST} mean = 0·01 and an S.D. prior = 0·05. Optimal K was determined *via* the δ-K likelihood method (Evanno *et al.*, 2005). GENECLASS2 (http://www1.montpellier.inra.fr/URLB/geneclass/geneclass.html/; Piry *et al.*, 2004) assessed self-assignment of individuals to population samples, using a simulated population size of 10 000 individuals per location and a 0·01 rejection level (Cornuet *et al.*, 1999).

ANALYSES FOR BOTH GENOMES

Numbers of private (unique) haplotypes (N_{PH}) and private alleles (N_{PA}), *i.e.* those occurring only in a single spawning group, were determined with CONVERT 1.31 (Glaubitz, 2004). F-statistics and their associated levels of significance were compared with FSTAT 2.9.3.2 (http://www2.unil.ch/popgen/softwares/fstat.htm/; Goudet, 2002) to discern whether population samples differed genetically using pair-wise θ_{ST} (Weir & Cockerham, 1984) and exact G non-parametric comparisons; both were executed in GENEPOP 4.2 (http://kimura.univ-montp2.fr/~rousset/Genepop.htm/; Raymond & Rousset, 1995; Rousset, 2008). Exact G-test comparisons were performed since they are non-parametric, have fewer assumptions and are less sensitive to sample-size effects. The F-based tests aided comparisons with levels of variation from other population genetic studies. Significance-associated probabilities of both types of tests were adjusted *via* sequential Bonferroni corrections to minimize type I errors (Rice, 1989).

To further delineate patterns among populations, the relationships between genetic distances [$\theta_{ST}\ (1 - \theta_{ST})^{-1}$] and geographic distances were evaluated separately for both genomic marker sets using the ln of geographic distances, measured as the shortest waterway distances (km). Regression significance was interpreted from 1000 GENEPOP permutations. Relationships between adjacent sampling sites additionally were assessed with Barrier 2.2 (http://ecoanthropologie.mnhn.fr/software/barrier.html/; Manni *et al.*, 2004) to identify genetically discontinuous assemblages, independent from *a priori* knowledge of relationships. The

barriers were ranked by bootstrap support from 2000 iterations in GENELAND 4.0 (http://www2.imm.dtu.dk/~gigu/Geneland/; Guillot *et al.*, 2005*a*, *b*; Guillot, 2008) using R 2.13.1 (www.r-project.org). Barriers with 50% bootstrap support and a majority of loci supporting were reported.

Hierarchical partitioning of genetic variation among possible grouping scenarios was tested with analysis of molecular variance (AMOVA; Excoffier *et al.*, 1992) in ARLEQUIN 3.5.1.3 (Excoffier & Lischer, 2010). Alternatives that were evaluated included distribution of variation among the population groups located in either lakes or rivers and between sites within *v*. outside of the Great Lakes drainage.

RESULTS

GENETIC DIVERSITY

All microsatellite loci were found to be unlinked and genotype frequencies conformed to HWE expectations after Bonferroni correction. There were 28 *cytb* haplotypes (mean ± S.E. $H_D = 0{\cdot}50 \pm 0{\cdot}06$) and 478 microsatellite alleles (mean ± S.E. $H_O = 0{\cdot}46 \pm 0{\cdot}03$) among *M. dolomieu* from the 18 population sampling sites (Table I). Mean values per sampling location were 3·4 haplotypes (range = 1–7) and 26·6 microsatellite alleles (range = 14–38; Table I).

Haplotype 1 was the most common in *M. dolomieu* (28% overall), occurring at sampling sites G–R [ranging from Georgian Bay in Lake Huron (site G) through to Paint Creek in the Ohio River (site R)], followed by haplotype 2 (14% overall) that was observed at sites A, G–I and L–O [Fig. 2 and Table SI (Supporting Information)]. All haplotypes were closely related, diverging from one another by a mean of 3·94 nucleotides (range = 1–8). No haplotypes were separated by more than four nucleotide positions from the most frequent haplotype 1. Just four haplotypes varied by four substitutions from haplotype 1; two of these were located in the upper Mississippi drainage and were closely related to one another (haplotypes 7 and 20), whereas the other two coincided with haplotype 1 in Cattaraugus Creek from eastern Lake Erie (site L, haplotype 19) and in Sandusky Bay from central Lake Erie (site I, haplotype 25). Ten of the 18 population samples contained at least one private haplotype, with the highest proportion (0·60) located in the Cannon River (site A) of the upper Mississippi River system. The greatest number of haplotypes and haplotypic diversity comprised seven haplotypes and $H_D = 0{\cdot}85$ for the Grand River collection from eastern Lake Erie (site K). Other spawning populations with high haplotypic variability were Georgian Bay in Lake Huron (site G; $H_D = 0{\cdot}83$ and six haplotypes) and Saginaw Bay in Lake Huron (site F; $H_D = 0{\cdot}71$, four haplotypes).

Similarly, mean ± S.E. nuclear microsatellite heterozygosity (H_O) for all population samples was $0{\cdot}46 \pm 0{\cdot}03$, ranging from 0·15 in Little Moose Lake (site P) to 0·60 for St Louis Bay (site D). The highest proportions of private alleles (0·08) characterized the Chagrin River (site J) and St Croix River (site C). Other locations with private alleles included Canon River (site A), Apple River (site B), Anchor Bay (site H), Hudson River (site Q) and Paint Creek (site R). Proportions of full siblings inferred per site varied from none (sites A–D, F, I, J, L, N and R) to 0·67 (sites E and P), with mean ± S.E. = $0{\cdot}15 \pm 0{\cdot}05$ over the entire range (Table I). For the microsatellite data, high levels of relatedness (*i.e.* inbreeding) were inferred based on positive values from internal relatedness estimations for all population samples surveyed (Table I). F_{IS}

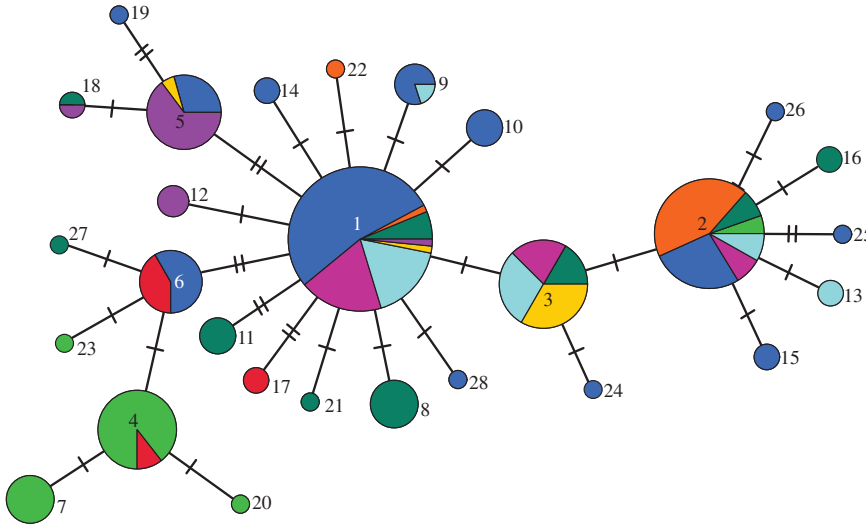

FIG. 2. Parsimony network among *Micropterus dolomieu* mtDNA *cytb* haplotypes (numbered) constructed using PopART (Leigh & Bryant, 2015). Circles are sized according to total observed relative frequency of each haplotype across all populations. ●, Upper Mississippi River; ●, Lake Superior; ●, Lake Huron; ●; Lake St Clair; ●, Lake Erie; ●, Lake Ontario; ●, St Lawrence River; ●, Hudson River; ●, Ohio River; —/—, unobserved mutational steps between observed haplotypes.

estimates showed a less uniform pattern, with some (but not all) population samples having positive values (again indicating possible inbreeding).

GENETIC DIVERGENCE

Genetic divergences among *M. dolomieu* populations across major regions were supported by all comparisons in both data sets (Table II). Fine-scale analyses found that most population samples were genetically distinct, with 67·3% of the *cytb* and 90·2% of the microsatellite comparisons being significantly different based on *F*-statistics (Table SII, Supporting Information). According to exact *G*-test comparisons, 75·0% of the *cytb* and 92·0% of the microsatellite comparisons significantly differed (Table SIII, Supporting Information). Of these comparisons, results from the *cytb* and microsatellite data sets were mostly congruent, with the *F*-statistics data set finding the same 99 comparisons significant (64·7%) after sequential Bonferroni correction and the *G*-test comparisons revealing the same 120 comparisons significant with correction (78·4%). Borderline and non-significant values may have been influenced by smaller sample sizes. The microsatellite data delineated 39 (25·5%) instances of significance where mtDNA did not, whereas mtDNA found four instances of significant differences (2·6%) where microsatellites did not.

PARTITIONING OF VARIANCE AMONG CATCHMENTS, LAKES, BASINS AND RIVERS

Occurrences and frequencies of mtDNA *cytb* haplotypes [Figs 2 and 3(a) and Table SI (Supporting Information)] often were distinctive among populations and regions.

TABLE II. Genetic divergence among *Micropterus dolomieu* populations measured by pair-wise (θ_{ST}) comparisons based on mtDNA *cytb* sequence variation (below diagonal) and eight nuclear DNA microsatellite loci (above diagonal). All comparisons were significant ($P < 0{\cdot}05$) before and following Bonferroni correction

	1	2	3	4	5	6	7	8	9
Upper Mississippi River	–	0·11	0·22	0·16	0·17	0·17	0·24	0·25	0·12
Lake Superior	0·29	–	0·09	0·23	0·10	0·22	0·19	0·26	0·15
Lake Huron	0·27	0·21	–	0·25	0·06	0·27	0·19	0·33	0·21
Lake Erie	0·55	0·61	0·38	–	0·21	0·18	0·26	0·34	0·19
Lake St Clair	0·29	0·20	0·11	0·38	–	0·15	0·13	0·25	0·15
Lake Ontario	0·40	0·38	0·16	0·42	0·05	–	0·25	0·38	0·19
St Lawrence River	0·35	0·31	0·09	0·44	0·05	0·02	–	0·31	0·18
Hudson River	0·49	0·48	0·23	0·72	0·32	0·38	0·21	–	0·20
Ohio River	0·45	0·42	0·28	0·64	0·24	0·40	0·36	0·51	–

Haplotype 1 was widespread (64/225, *i.e.* 28% of individuals), being prevalent throughout most of the Great Lakes drainage, as well as in the Hudson River (site Q) and Paint Creek of the Ohio River (site R). Haplotype 2 (32/225, *i.e.* 14% of individuals) was distributed sporadically across a wide geographic range throughout most of the Great Lakes drainage and as far west as the Cannon River (site A). Haplotype 4 was most common in the upper Mississippi River drainage and also occurred in Lake Superior. Haplotypes unique to specific regions included: 7, 20 and 23 in the Mississippi River Drainage, 17 in Lake Superior, 11,16, 21 and 27 in Lake Huron, 22 in Lake St Clair, 9, 10, 14, 15, 19, 24, 25, 26 and 28 in Lake Erie, 13 in the St Lawrence River and 12 in Paint Creek.

The parsimony network (Fig. 2) indicated the prevalence of haplotype 1, which was distributed from Georgian Bay in Lake Huron (site G) to Paint Creek in the Ohio River

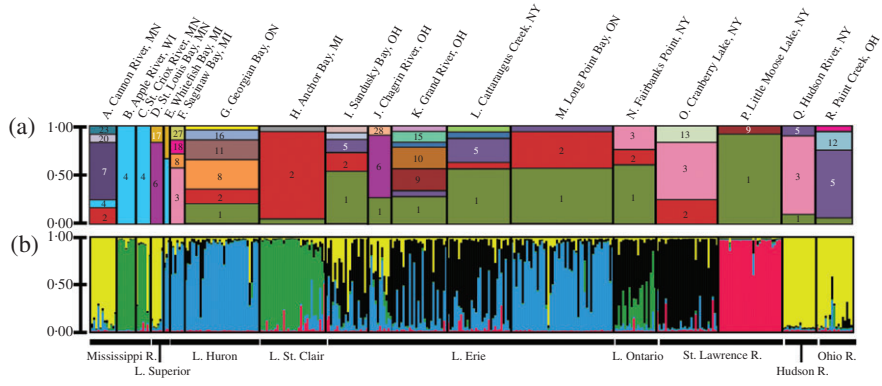

FIG. 3. *Micropterus dolomieu* population structure estimated by (a) mtDNA *cytb* haplotype frequencies representing one of the 28 haplotypes (numbered and color coded) and (b) Bayesian assignments using STRUCTURE 2.3.4 [$K = 5$, with the ΔK likelihood (Evanno *et al.*, 2005)] for eight microsatellite loci. Individuals are represented by thin vertical bars coloured according to inferred group membership. ——, separation between different spawning group populations.

(site R). Nine haplotypes (32% of haplotypes) varied by just single nucleotides from haplotype 1, including haplotypes 3, 9, 10, 14, 22 and 28, which also were the most abundant. The most genetically distant haplotypes (from haplotype 1) were 7, 19, 20 and 25, diverging by four nucleotide substitutions. The Mississippi River grouping (sites A–C, haplotypes 4, 7, 20 and 23) was the most defined cluster from haplotype 1, differing by three to four steps.

The Bayesian phylogenetic tree of the 28 haplotypes (Fig. 4) illustrated relatively close relationships among most haplotypes. There were three well-supported clades; one linked three haplotypes (5, 18 and 19) uniquely found in Paint Creek (site R), another contained six haplotypes (6, 7, 23, 4, 20 and 27) that mostly occurred in the upper Mississippi River and Lake Superior (sites A–F and J) and a third clade possessed six haplotypes (2, 13, 15, 16, 15 and 26) mostly from the lower Great Lakes (sites H–O). Haplotypes 7 and 23 from the Cannon River (site A) also grouped together as a well-supported clade.

Bayesian analyses of microsatellite data using STRUCTURE [Fig. 3(b)] identified $K = 5$, 9 and 14 population clusters (Figs S1 and S2, Supporting Information), with $K = 5$ being the optimal number determined by the delta K method (Evanno *et al.*, 2005). Neither microsatellite nor *cytb* sequences differentiated among adjacent sites in Lake Erie. Results from the GENECLASS analysis indicated high self-assignment at most spawning locations, with the exceptions of Georgian Bay (site G), Sandusky Bay (site I), Cattaraugus Creek (site L) and Cranberry Lake (site O), in which higher proportions of individuals assigned to other spawning groups (Table SIV, Supporting Information). Overall, however, these results were consistent with those found by STRUCTURE (Fig. S2, Supporting Information).

AMOVA partitioning of genetic variation supported significant primary divergence between the two population groupings from within *v*. outside the Great Lakes' drainage, along with distinctive differentiation among their component populations (Table III). There was no significant differentiation between population samples from lakes *v*. rivers (Table III).

Populations from outside the Great Lakes catchment were separated by the strongest barriers to gene flow inferred from Barrier analysis (Fig. 1). *Micropterus dolomieu* from the Hudson River (site Q) appeared very distinctive from other populations, having the most pronounced genetic barrier based on microsatellite evidence (I, six loci) and the second greatest barrier based on *cytb* evidence (*cytb* 2). The upper Mississippi River drainage (sites A–C) was denoted by the strongest *cytb* inferred barrier (*cytb* 1) from the other populations and the third greatest microsatellite inferred barrier (III, seven loci). The second largest microsatellite inferred barrier (II, six loci) divided the Cranberry Lake population (site O) from the others, a difference not shown by the *cytb* sequences. Within the Great Lakes, the strongest division was between populations from the upper (Superior, Michigan, Huron, St Clair; sites D–H) *v*. the lower Great Lakes (Erie, Ontario; sites I–N), which was recovered with both genomes (IV, six loci and *cytb* 3). The upper Mississippi River populations were subdivided further, with Cannon River (site A) distinguished by the fifth strongest *cytb* inferred barrier (*cytb* 5) and Cannon River and Apple Creek (sites A and B) characterized by the fifth largest microsatellite inferred barrier (V, five loci). The Lake St Clair population was divided from the other upper Great Lakes (VI, seven loci and *cytb* 4). The remaining barriers separated out the St Louis Bay (site D; VII, six loci and *cytb* 6), Ohio River (site R; *cytb* 7), St Lawrence River (sites O and P; VIII, five loci and *cytb* 9), Whitefish Bay (site E;

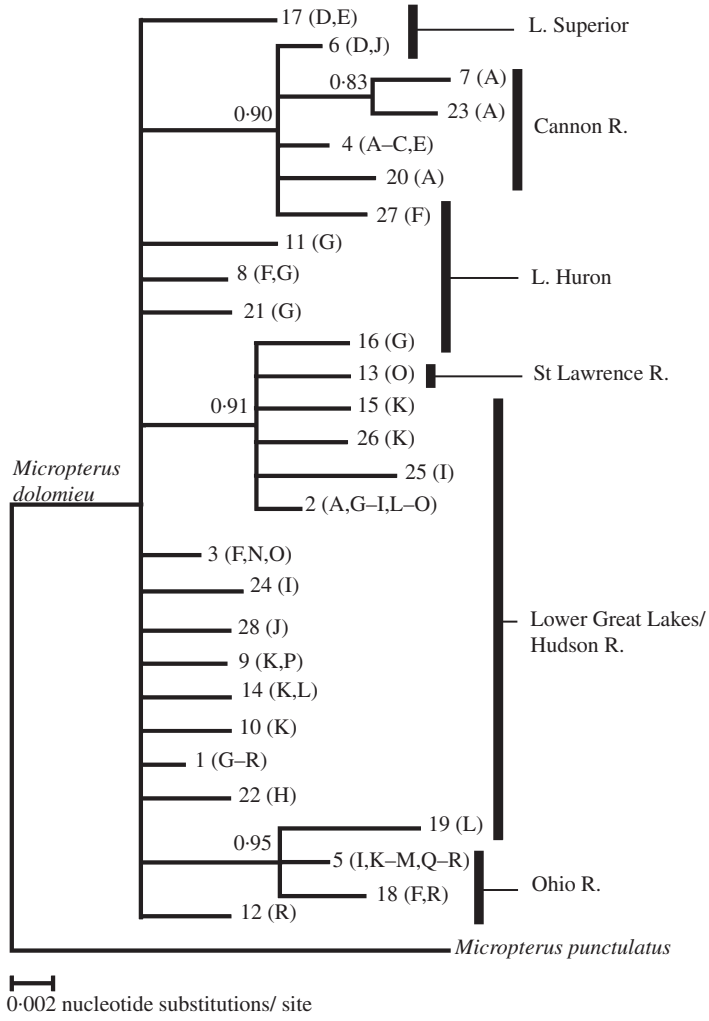

FIG. 4. Phylogenetic relationships among 28 mtDNA cytochrome *b* haplotypes of *Micropterus dolomieu* from the Bayesian 50% majority-rule consensus tree, rooted to *Micropterus punctulatus*, inferred using MRBAYES 3.2.2 (Huelsenbeck & Ronquist, 2001; Ronquist *et al*., 2011). Letters in parentheses indicate locations where the haplotype was recovered, and ___ denote major geographic regions. Values above nodes are Bayesian posterior probabilities (percentage values above 0·50 are shown).

X, six loci, *cytb* 8), Lake Ontario (site N; IX, five loci) and Sandusky Bay populations (site I; *cytb* 10) (Table IV).

ISOLATION BY GEOGRAPHIC DISTANCE

Relationships among *M. dolomieu* populations followed a pattern of genetic isolation by geographic distance, with both the mtDNA and the microsatellite data sets (Fig. 5). Although the largest genetic differences between populations matched those with large geographic demarcations, there were also a number of sampling sites separated

TABLE III. Results of AMOVA based on cyt *b* sequences and eight microsatellite loci from *Micropterus dolomieu* populations [A–R (see Fig. 1)]

	Cytb				Microsatellites			
Source of variation	Variation (%)	Φ	*P*	Mean F_{ST}	Variation (%)	Φ	*P*	Mean F_{ST}
Great Lakes (sites D–P) *v.* outside of Great Lakes drainage (A–C, Q, R)	18·8	0·18	<0·001	0·44	9·6	0·10	<0·001	0·27
Among the sampling sites within the two groups (inside *v.* outside)	27·2	0·45	<0·001	0·49	18·9	0·21	<0·001	0·23
Within sampling sites	54·0	0·33	<0·001	–	71·5	0·29	<0·001	–
Between lake (D–I, M–P) and river (A–C, J–L, Q, R) groupings	2·6	0·03	>0·05	0·41	0·85	0·00	>0·05	0·23
Among the sampling sites within the two groups	37·6	0·39	<0·001	0·41	24·2	0·24	<0·001	0·23
Within sampling sites	59·8	0·40	<0·001	–	76·7	0·23	<0·001	–

by large geographic distances that exhibited little to no genetic difference. Many higher-than-average pair-wise comparisons for the *cytb* data were associated with sites B (Apple Creek) and C (St Croix River) for eight of 10 comparisons and with site P (Little Moose Lake) for four of 10 comparisons [Fig. 5(a)]. All 11 higher-than-average pair-wise comparisons for microsatellites were associated with site P [Fig. 5(b)].

DISCUSSION

GENETIC DIFFERENTIATION AND RELATION TO QUESTIONS

This investigation discerns strong patterns of differentiation among spawning groups of *M. dolomieu* from 18 population sites across the North American Great Lakes and

TABLE IV. GENECLASS *Micropterus dolomieu* population assignments (as proportions of total number of individuals) from microsatellite data

	1	2	3	4	5	6	7	8	9
Upper Mississippi River	1·00*	–	–	–	–	–	–	–	–
Lake Superior	–	1·00*	–	–	–	–	–	–	–
Lake Huron	0·02	–	0·74*	–	0·21	–	–	–	0·02
Lake Erie	–	–	–	1·00*	–	–	–	–	–
Lake St Clair	–	–	0·08	–	0·88*	–	0·01	–	0·03
Lake Ontario	–	–	–	–	0·25	0·75*	–	–	–
St Lawrence River	–	–	0·02	–	0·28	–	0·69*	–	0·02
Huron River	–	–	–	–	–	–	–	1·00*	–
Ohio River	–	–	–	–	–	–	–	–	1·00*

*Self-assignments within catchments.

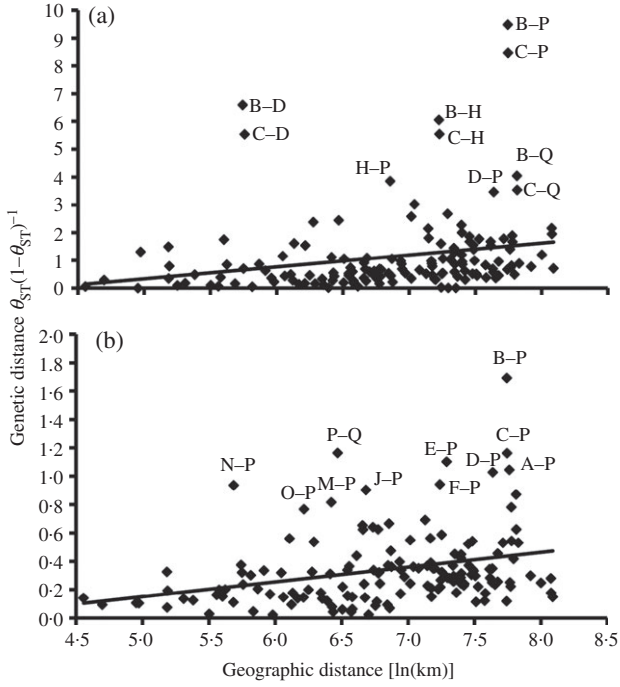

FIG. 5. Genetic distances [θ_{ST} $(1-\theta_{ST})^{-1}$] plotted against the ln of geographic distances (km) for (a) mtDNA haplotype data ($y = 0{\cdot}43x - 1{\cdot}81$, $R^2 = 0{\cdot}05$, $P < 0{\cdot}001$) and (b) nuclear DNA microsatellite data ($y = 10{\cdot}00x - 0{\cdot}37$, $R^2 = 0{\cdot}1$, $P < 0{\cdot}01$). Outlier population pairs (*e.g.* B–D, as denoted in Fig. 1) are shown.

adjacent areas, based on both mitochondrial and nuclear DNA data. These results help to answer the original questions.

Do similar levels of genetic diversity characterize M. dolomieu *populations with both genomes?*

Results demonstrate that overall levels of genetic diversity are very similar between the two genomes. High nuclear microsatellite diversity, however, did not always coincide with high mtDNA variability and *visa versa*. Notably, the population from Georgian Bay in Lake Huron (site G) possessed high mtDNA diversity (0·83) yet had modest microsatellite variability (0·38), whereas population samples from Apple Creek (B) and the St Croix River (C) in the upper Mississippi River system were monomorphic in mtDNA (0·00), yet contained average levels of nuclear microsatellite variability (0·49 and 0·55, respectively). It is possible that this discord reflects sample-size effects or the respective evolutionary mutational scales of mitochondrial *v*. nuclear DNA genomes, which would provide an intriguing avenue for further investigation.

Are patterns of genetic divergence similar between the two M. dolomieu *genomes?*

Both genomes do indicate that populations from all major geographic regions diverge significantly from each other (Tables II and III) and almost all local population samples also differ significantly (Tables SI–SIV, Supporting Information). More fine-scale

differences were resolved with the microsatellite results, probably due to their having multiple loci and more alleles and consequently greater discrimination power. Results from the mtDNA analysis probably reflect population origins that trace to historical expansion from glacial refugia rather than recent genetic differentiation. Nevertheless, both data sets reveal congruent patterns.

Is genetic variation significantly partitioned among catchments, rivers, lakes and basins?

Results do indeed indicate that the strongest barriers to gene flow separate populations outside of the Great Lakes from those within. Significant divergences among many population sites distinguish *M dolomieu* from river systems, lakes and basins. Differential occurrence and distribution of recovered mitochondrial haplotypes denote indigenous population groups from lakes, rivers and geographic regions that may reflect the influences of their historical glacial refugia origins and colonization pathways.

Are patterns of differentiation related to genetic isolation by geographic distance using both genomes?

Both data sets support the hypothesis of differentiation with geographic distance. This indicates that geographic demarcation among population sites significantly influenced the genetic divergence among spawning groups. As this correlation was relatively weak, however, colonization from glacial refugia also probably affected the patterns of relationships among populations.

COMPARISONS WITH DIVERSITY OF OTHER GREAT LAKES FISHES

Micropterus dolomieu populations in the Great Lakes (mean ± S.E. $H_O = 0{\cdot}46 \pm 0{\cdot}03$; range = 0·15–0·60) display lower observed microsatellite heterozygosity than did some other popular fishery species with overlapping geographic ranges, such as *P. flavescens* (mean ± S.E. $H_O = 0{\cdot}53 \pm 0{\cdot}01$, range = 0·39–0·63; Sepulveda-Villet & Stepien, 2012; Stepien *et al.*, 2015*a*, *b*) and *S. vitreus* (mean ± S.E. $H_O = 0{\cdot}68 \pm 0{\cdot}03$, range = 0·52–0·77; Haponski & Stepien, 2014; Stepien *et al.*, 2015*b*). This may be related to smaller contemporary spawning population sizes and less genetic migration of *M. dolomieu* individuals, leading to more apparent inbreeding and reduced genetic variability *via* drift. mtDNA haplotypic diversity of *M. dolomieu* (mean ± S.E. $H_D = 0{\cdot}50 \pm 0{\cdot}06$, range = 0–0·85) is almost as high as *S. vitreus* (mean ± S.E. $H_D = 0{\cdot}53 \pm 0{\cdot}02$, range = 0·15–0·88), with both being considerably greater than for *P. flavescens* (mean ± S.E. $H_D = 0{\cdot}31 \pm 0{\cdot}05$, range = 0–0·82). Each of these three species possessed a single dominant haplotype in the Great Lakes. This pattern probably resulted from their common phylogeographic and glacial colonization histories and it is interesting that all three species appear to have evolved similarly in response to these historical events.

Proportions of full siblings in *M. dolomieu* samples (mean ± S.E. $p_S = 0{\cdot}15 \pm 0{\cdot}05$, range = 0–0·67) were similar to those for *P. flavescens* (mean ± S.E. $p_S = 0{\cdot}16 \pm 0{\cdot}02$, range = 0–0·38; Sullivan & Stepien, 2015) and about twice those characterizing *S. vitreus* populations (mean ± S.E. $p_S = 0{\cdot}08 \pm 0{\cdot}02$, range = 0–0·76; Haponski & Stepien, 2014). This appears to indicate a tendency for proximate fish to be members of family groups and may reflect how limited overall lifetime migration patterns affect gene

flow among locales. This is supported by the consistent positive internal relatedness (IR) values estimated for each population sample. The moderate to high relatedness in these sampled populations might indicate inbreeding, although significant loss of diversity does not seem to have occurred. *Micropterus dolomieu* appears to disperse the least of these three species, having a maximum lifetime dispersal reported as 10 km (Lyons & Kanehl, 2002). In comparison, *P. flavescens* may migrate up to 48 km (with the occasional individual travelling as far as 200 km; Rawson, 1980) and *S. vitreus* up to 300 km (Colby *et al.*, 1979). A smaller lifetime dispersal distance probably lessens gene flow among regions and may increase inbreeding among localized populations. For example, Ridgway *et al.* (1991) discovered that adult male *M. dolomieu* displayed high breeding site fidelity, with 81% of the males building nests within 200 m of their previous year's nest site. Since nest locations were stable, lack of migration by males may limit gene flow and preserve private alleles within a given population. Similarly, Gross & Kapuscinski (1997) discerned that over 50% of spawned individuals were produced by <10% of available males in Lake Opeongo, Ontario, Canada. It is possible the pronounced success of few individuals could contribute to the high degree of relatedness detected in results presented here. As discussed by Bouzat (2010), increased inbreeding often is associated with lower genetic diversity; but this may not always be the case. *Micropterus dolomieu* may use cues (*e.g.* olfactory and visual) to distinguish close relatives and avoid inbreeding, thereby reducing the possible negative effects of inbreeding depression. For example, such sensory clues appear to allow the perch *Perca fluviatilis* L. 1758 to identify kin from non-relatives and circumvent inbreeding (Behrmann-Godel *et al.*, 2006; Behrmann-Godel & Gerlach, 2008). Whether such factors also may occur in *M. dolomieu* may be important for the long-term sustainability of population groups and should be investigated further.

COMPARISONS WITH THE DIVERGENCE PATTERNS OF OTHER GREAT LAKES FISHES

Populations of *M. dolomieu* display evidence for origins from the three glacial refugia (Missourian, Mississippian and Atlantic) in this study, which also characterized *S. vitreus* (Stepien *et al.*, 2009, 2015*a*, *b*; Haponski & Stepien, 2014), *P. flavescens* (Sepulveda-Villet & Stepien, 2012; Sullivan & Stepien, 2014, 2015) and several other Great Lakes fishes (April *et al.*, 2013). Those refugia provided retreat from climatic, habitat and geophysical changes that occurred during the Pleistocene glaciations (Hewitt, 2004; Provan & Bennett, 2008). The upper Mississippi River populations of *M. dolomieu* traced to the Missourian refugium, as likewise was detected for *A. nebulosus* of western Lake Superior (Murdoch & Hebert, 1997). The majority of all Great Lakes *M. dolomieu* populations appear to have originated from the Mississippian refugium, as was the case for most Great Lakes' species (Mandrak & Crossman, 1992; Todd & Hatcher, 1993). Eastern populations of *M. dolomieu* from the St Lawrence and Hudson Rivers traced their colonization to the Atlantic refugium, which provided protection in diverse habitats east of the Appalachian Mountains (Bernatchez, 1997). Similar to findings for *P. flavescens* by Sepulveda-Villet & Stepien (2012), haplotype 1 was probably widespread prior to the glaciations, leading to its currently ubiquitous distribution in sites that were colonized from both the Mississippian and Atlantic refugia. These parallel broad-scale patterns for *M. dolomieu*, *S. vitreus* and *P. flavescens* indicate their shared phylogeographic history (*i.e.* congruent glacial refugia

and colonization pathways) and point to the importance of maintaining these genetic demarcations among key population groups for conservation planning to protect the natural population structure of these exploited fishery species.

The present study discerns that large genetic breaks distinguish populations outside the Great Lakes from those within and delineate those in the upper from the lower Great Lakes. This pattern is similar to that recovered for *P. flavescens* (Sepulveda-Villet & Stepien, 2012), but varies from *S. vitreus* (Haponski & Stepien, 2014). The latter possessed many barriers separating populations outside of the Great Lakes, as well as several within them, but lacked the demarcation between the upper and the lower Great Lakes. This is probably due to differences in migration patterns and spawning-site fidelity among the three species.

RELEVANCE TO OTHER *M. DOLOMIEU* STUDIES

Although *M. dolomieu* is an economically and ecologically important species in the Great Lakes and elsewhere, its genetic diversity and broad and fine-scale population relationships are relatively understudied. Borden & Stepien (2006) described moderate variability and some genetic divergence among groups using partial mtDNA sequences and microsatellite loci, but were limited by low-resolution power and small sample sizes. Stepien *et al.* (2007) examined broad-scale patterns using eight microsatellite loci and larger samples, discerning differences among the Great Lakes populations, as well as for river *v.* lake sites. Bagley *et al.* (2011) examined mtDNA *cytb* haplotypes of populations in Arkansas and southern Missouri, U.S.A. That study recovered 17 distinct haplotypes (three of which were shared with the present analysis: haplotypes 3–5). Hallerman *et al.* (2015) discovered that *M. dolomieu* from the Tennessee River drainage were markedly differentiated from populations in the New and James River drainages, which may be related to different glacial refugia origins (Mississippian *v.* Highlands) as well as artefacts of past stocking supplementation.

The present investigation indicates that *M. dolomieu* possesses moderately high genetic diversity, strong broad-scale patterns related to post-glacial colonization and pronounced fine-scale population divergence, which are probably due to a combination of genetic isolation by geographic distance, limited migration and spawning-site fidelity. These results, coupled with those of earlier studies, indicate that *M. dolomieu* population structure and genetic diversity stem from interplay between phylogeographic history and contemporary processes that are in many ways similar to other North American fish species (Stepien *et al.*, 2009, 2015*a*, *b*; Sepulveda-Villet & Stepien, 2012; Haponski & Stepien, 2014; Sullivan & Stepien, 2014, 2015).

In conclusion, this investigation identifies a number of important aspects concerning the broad and fine-scale population genetic patterns of *M. dolomieu* spawning groups. First, moderate genetic diversity and strong population genetic divergence characterize both genomes. Broad-scale population genetic patterns of *M. dolomieu* indicate the probable influence of post-glacial recolonization patterns tracing to Missourian, Mississippian and Atlantic refugia. Robust fine-scale genetic patterns are discerned among population groups, which are related to a combination of factors, including geographic demarcation, limited dispersal and spawning-site fidelity. These results denote the importance of both historical and contemporary influences on population structure in an exploited fishery species. Lastly, this enhanced understanding of the contemporary influences on population structure, specifically the interplay

among behavioural philopatry, inbreeding and genetic diversity may provide useful information to help manage the demographic variability and viability of *M. dolomieu* effectively.

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Supporting Information

Supporting Information may be found in the online version of this paper:

Table SI. Frequencies of *Micropterus dolomieu* population sample mtDNA *cytb* sequence haplotypes.

Table SII. Genetic divergence patterns among smallmouth bass *Micropterus dolomieu* population samples (see Fig. 1 for locations) measured by pair-wise comparisons (θ_{ST}) based on mtDNA *cytb* sequence variation (below diagonal) and on eight nuclear DNA microsatellite loci (above diagonal). Values with black backgrounds remained significant upon sequential Bonferroni correction, and values with grey background were significant prior to sequential Bonferroni correction, but not after. Values shown on white were not significant.

Table SIII. Exact *G*-test values among all locations (A–R, see Fig. 1) based on eight nuclear microsatellite loci (above diagonal) and mitochondrial DNA *cytb* sequences (below diagonal). Black background indicates significant values ($P < 0{\cdot}05$); white background is not significant.

Table SIV. Results of Bayesian assignments to *Micropterus dolomieu* population samples A–R (see Fig. 1) based on eight microsatellite loci using GENECLASS2. Underlined values denote the percentage of individuals that self-assigned to source collection.

Fig. S1. Results for each possible number of K clusters based on STRUCTURE HARVESTER (Earl & von Holdt, 2012) are shown for ΔK, indicating greatest support for $K = 5$.

Fig. S2. Smallmouth bass *Micropterus dolomieu* population samples A–R (see Fig. 1) estimates from Bayesian assignments using STRUCTURE 2.3.4, (a) $K = 9$ and (b) $K = 14$ for eight microsatellite loci. Individuals are represented by thin vertical bars coloured according to inferred group membership. Vertical black lines separate different population samples (spawning groups).

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